

Hazardous-area CCTV. High-level production visibility.

Engineered monitoring for elevated process areas, corrosive environments and low-light production operations - designed, implemented and commissioned by D-TECH.

D-TECH DELIVERY

SURVEY -> DESIGN -> SAFE INSTALLATION -> MONITOR



ORIGINAL ORDER EVIDENCE

A two-phase production-monitoring deployment at GNFC Bharuch.

The 2016 base system was followed by a 2017 camera expansion order.

2016

base system

Neem Coating Urea Plant

2017

expansion

Additional fixed IR dome

3

verified cameras

2 anti-corrosion + 1 dome

12 TB

recording

Two 6 TB hard disks

Rs 3.78 L

combined material

INR 377,622 excluding tax

VERIFIED SCOPE

Supply, installation, testing and commissioning of IP video surveillance.

Base PO MM/KKV/4500053197 references RFP/CCTV/GNFC/NCP/2016; expansion PO MM/KKV/4500057935 is dated 09 August 2017.

PRODUCTION MONITORING CHALLENGE

Process visibility must remain reliable in a demanding plant environment.

High structures, corrosive atmosphere, mixed lighting and hazardous zoning change how CCTV must be engineered.



Elevated viewpoints

Production towers and process corridors require high-level overview cameras.

Corrosive exposure

Camera construction, supports, glands and pathways must resist the environment.

Low-light operation

Night production needs usable detail without excessive image noise.

Safety-controlled work

Installation requires permits, access planning and work-at-height controls.

SOLUTION DRAWING 01

Elevated cameras create a layered production-monitoring view.

The design combines process overview, equipment context and access-route visibility.



SOLUTION DRAWING 02

Hazardous-area engineering separates the classified zone from safe services.

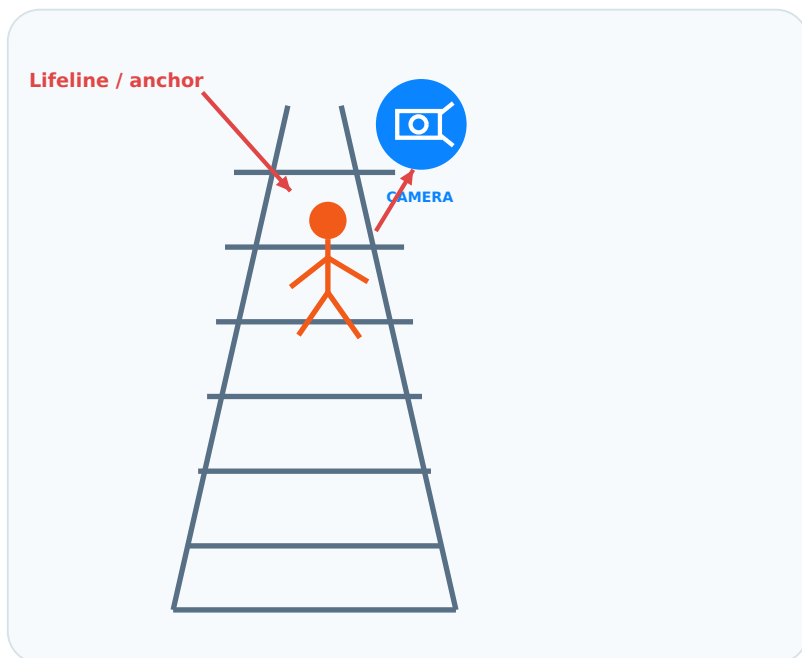
Final camera and accessory certification must match GNFC-approved zone classification, gas group and temperature class.



SOLUTION DRAWING 03

Work-at-height installation is engineered as carefully as the camera view.

Safe access, anchorage, tools, exclusion zones and commissioning checks are planned before work begins.



01

Permit to work

Approved method, zone clearance and task-specific risk assessment.

02

Safe access

Certified scaffold / manlift with inspected platform and access route.

03

Fall protection

Full-body harness, double lanyard, lifeline and verified anchorage.

04

Area control

Barricade, ground spotter, tool tethering and exclusion zone.

05

Gas / process control

Testing, shutdown coordination and GNFC operations clearance.

06

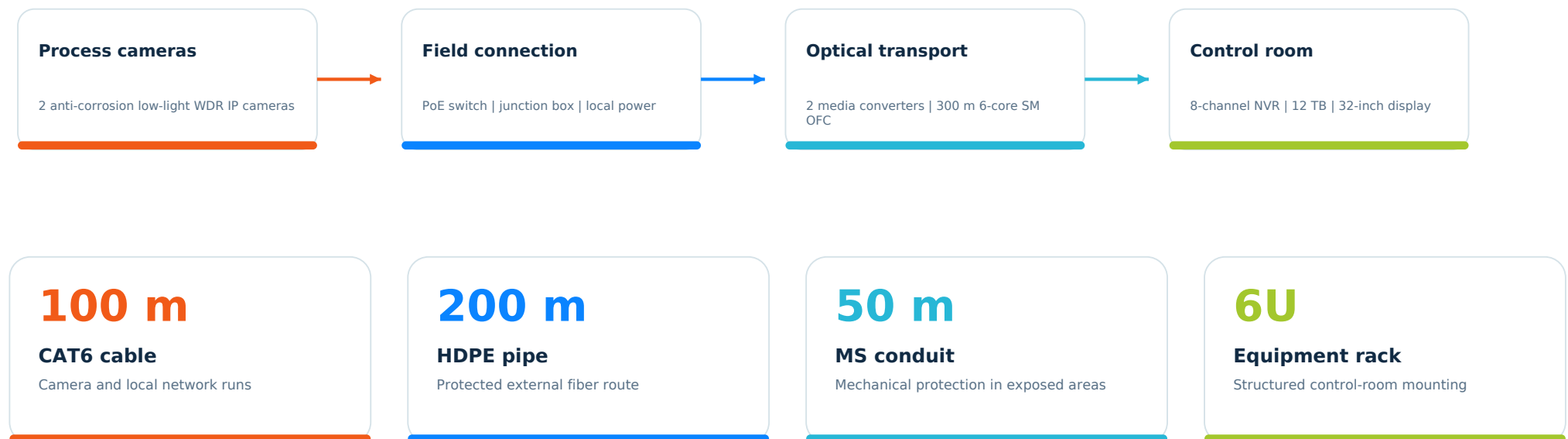
Two-stage QA

Mechanical check at height followed by video validation below.

SOLUTION DRAWING 04

Fiber transport protects production video across plant distance.

The verified 2016 scope combines field power, CAT6, single-mode OFC, media conversion and centralized recording.



SECURITY OPERATIONS

Production video becomes actionable evidence in the control room.

Operators receive live views, low-light context, recorded history and controlled incident export.

Live production view

Elevated process cameras maintain overview of critical operating areas.

Low-light monitoring

Ultra-low-light imaging supports night operations and reduced illumination.

WDR scene balance

High-contrast process lighting remains readable across bright and dark zones.

Incident playback

NVR search and synchronized playback reconstruct the event sequence.

Evidence retention

Two 6 TB disks provide 12 TB of ordered recording capacity.

Expansion path

The 2017 fixed dome and PoE switch extend coverage without redesigning the base.

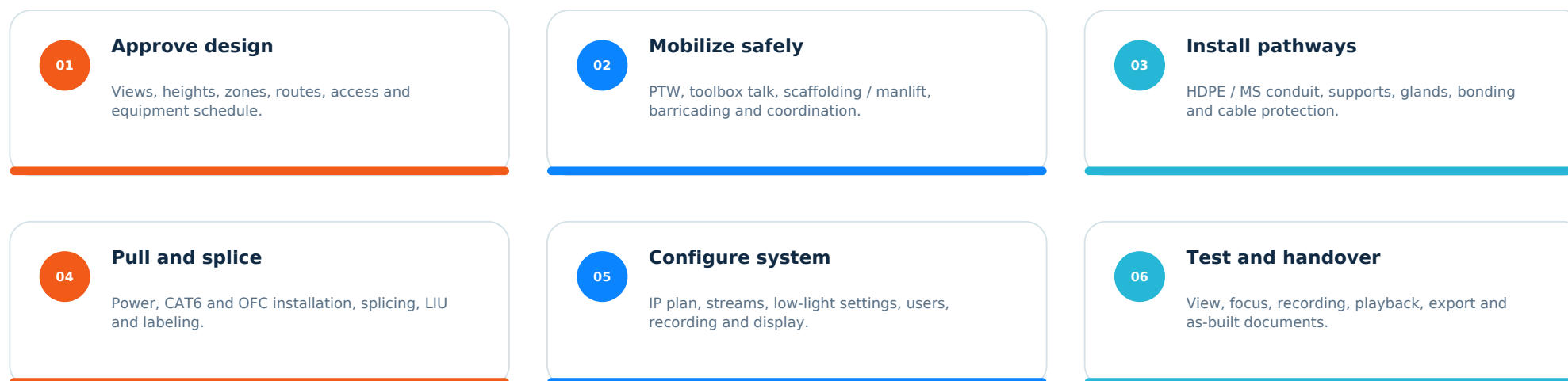
OPERATIONAL RESULT

Faster awareness | safer verification | traceable evidence | accountable response

IMPLEMENTATION

A permit-led deployment sequence protects plant safety and system quality.

The project moves from approved drawings to safe installation, configuration, testing and as-built handover.



QUALITY GATE

No elevated camera is accepted until safe mounting, stable transport and usable production video are verified.

TESTING AND ACCEPTANCE

Performance is verified from the camera mount to recorded evidence.

The QA plan confirms physical integrity, optical transport, image quality, recording and documentation.

Mechanical QA

Mounting, corrosion protection, glands, supports, conduit and bonding.

Height-work closure

Platform clearance, tool count, barricade removal and permit closure.

Fiber QA

Splice quality, LIU mapping, media conversion and link stability.

Video QA

View, focus, low-light image, WDR balance and scene coverage.

Recording QA

Continuous recording, storage health, playback, search and export.

Documentation

Camera list, IP schedule, fiber route, test sheets and as-built drawings.

ACCEPTANCE RESULT

A safe, serviceable and evidence-ready production-monitoring platform.

BUSINESS IMPACT

High-level production visibility without compromising engineering discipline.

The system links harsh-environment camera selection, fiber transport and safe installation into one accountable solution.

Production awareness

Elevated views improve context across major process structures.

Safer verification

Operators can assess conditions before dispatching personnel.

Night visibility

Low-light technology maintains monitoring beyond daylight hours.

Corrosion resilience

Anti-corrosion camera construction supports the process environment.

Evidence continuity

12 TB recording supports review, investigation and learning.

Scalable foundation

Fiber and PoE architecture support later camera expansion.

PROJECT RECORD

PO 4500053197 (2016) + PO 4500057935 (2017) | GNFC Bharuch | D-TECH implementation.



CASE STUDY CONCLUSION

Designed for the zone. Installed for the height. Monitored for production.

D-TECH delivered a production-monitoring foundation built around harsh-environment camera selection, safe work-at-height execution, fiber connectivity and reliable recording.

D-TECH SOLUTION INTEGRATORS PVT. LTD.

Industrial CCTV | hazardous-area design | work-at-height installation | lifecycle support

